The last fixed-point budgets at odd degree

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le \mathrm{Alt}(\Omega)$ be transitive with a connected string C-marking and all maximal omissions intransitive. Assume $r \ge 6$, $n = 2r+1$ and the actual transitive six-point head on $D = \{1, \dots, 6\}$, with the literal interface

$$\tau = \rho_3 = (6\ b)\delta, \quad R = G_{\ge 3} = \mathrm{Alt}(\Omega \setminus \{1, \dots, 5\}), \quad \sigma = \rho_4, \quad \sigma|_D = 1.$$

The odd involution δ fixes D, b. Put $H = G_{\ge 5}$ and $K = G_{>5}$, with H fixing D, b. Suppose each tail colour has two full ambient-eligible edges in one actual H-orbit, and normalize these assignments by the acquired whole-orbit transport.

Theorem 1. *Under these hypotheses H cannot have nine or more fixed points.*

Proof. Write $f = |\mathrm{Fix}(H)|$, and let $c \ge 1$ count its nontrivial orbits. The complete ambient interval theorem applies to all $r-5$ tail colours and gives

$$2(r-5) \le n - f - c, \qquad f + c \le 11. \tag{1}$$

Thus $f \ge 9$ leaves only $f = 9, c \le 2$ or $f = 10, c = 1$. The acquired actual head argument gives

$$x = b^\sigma \ne b, \qquad \delta x = x, \qquad Kx = x. \tag{2}$$

Moreover x is not fixed by H. If it were, H would fix $D \cup \{b, x\}$ pointwise. The prefix $G_{\le 4}$ preserves that set: the head preserves D, τ fixes x and exchanges $6, b$, and σ fixes D and exchanges b, x. The full group would preserve its eight points, contrary to transitivity and $n \ge 13$.

Suppose first that $c = 1$. The involution δ preserves the unique nontrivial H-orbit and fixes its point x, so centralization makes it trivial on that orbit. Outside D, b there are at most three H-fixed points. As δ is odd, it is literally one transposition on those points. The complete acquired exterior-pair obstruction applies with every required ambient eligible pair, and excludes this case.

We may therefore assume $f = 9, c = 2$. For the two orbits put $d_i = |F_i| - |O_i|/2$. The normalized bounds give

$$d_1 + d_2 = -1, \qquad d_p \le 0, \qquad d_i < 0 \quad (i \ne p), \tag{3}$$

where $5 \in F_p$. The involution δ preserves both orbits. Otherwise they have equal size m; transporting their assigned pairs by τ and using the selected-colour bound gives combined deficit at most $-\lceil m/2 \rceil \le -2$ for $m \ge 3$. For $m = 2$ both assignments are empty and the deficit is exactly -2. Either conclusion contradicts (3).

If δ fixes both orbits pointwise, its odd action is again the single transposition of the two additional fixed points, already excluded. Otherwise let W be an orbit with nontrivial internal action. It is free there and has even size $2q$. Its deficit is at most -1: use the complete pair-forest bound when its assignment is nonempty, and its exact deficit otherwise. It cannot be the first assigned orbit, since the other deficit is negative. Equation (3) therefore forces

$$d_W = -1, \qquad d_V = 0, \qquad V = O_p.$$

The involution δ fixes V pointwise; otherwise the same bound would make $d_V \le -1$. The point x lies in V, by (2) and the preceding nonfixation argument. Every possible further action of δ is on the two additional H-fixed points.

If $q = 1$, its assignment is empty, and oddness forces δ to fix those two extra points. It is the literal transposition on W, again impossible. If $q \geq 3$, then F_W has $q - 1$ colours, all at least six. Its normalized K-action is intransitive. The complete odd-degree row theorem gives a last assigned label $t < r - 1$ with $|\operatorname{Fix}(G_{>t})| \geq 2t + 3$. The remaining nonempty paired interval has support at least $2(r - 1 - t) + 1$, forcing $n \geq 2r + 2$. Thus only $q = 2$ remains.

In this case $|W| = 4$, $F_W = \{j\}$ with $j \geq 6$, and F_V is the complementary set of tail colours. As $|V| = 2|F_V|$ and K fixes $x \in V$, the complete fixed-anchor lemma makes F_V an initial interval. Its unique missing label must therefore be $j = r - 1$. Put $Q = \langle \rho_j \rangle$.

We give the exact two-row argument with the present fixed points. The selected eligible j-edge joins distinct δ-pairs in W. No other tail mark can exchange those pair vertices: it and τ would then connect all of W in G_j, contrary to eligibility. Thus K is transitive on the two pair vertices. Normalization makes it intransitive on W, so it has two orbits of size two, each meeting each pair once. Its pair-trivial elements fix both rows pointwise. Consequently $\rho_j|_W$ is a double transposition and every other mark of K fixes W pointwise.

For $4 \leq l \leq j - 2$ put $Z_l = W^{G_{\leq l}}$. Each is a union of nontrivial two-point Q-orbits, since the prefix centralizes Q. We have $|Z_4| \geq 6$: otherwise $G_{\leq 4}$ and H would both preserve W, contradicting transitivity. Every Z_l is disjoint from the ten Q-fixed points

$$\operatorname{Fix}(H) \cup \{x\}.$$

For $4 \leq l \leq j - 3$, every mark of index at least $l + 2$ except ρ_j fixes W and commutes with the prefix, so fixes Z_l. The group Q and the prefix preserve Z_l. Transitivity forces ρ_{l+1} to enlarge this proper set, by at least one whole two-point Q-orbit. Therefore

$$|Z_{j-2}| \geq 6 + 2(j - 6) = 2j - 6, \qquad n \geq (2j - 6) + 10 = 2j + 4.$$

But $j = r - 1$ gives $n = 2j + 3$, the final contradiction. □

Attribution and boundary. The complete interval bound, normalized deficit theorem, fixed-anchor theorem, odd-degree row exit and literal exterior-pair obstruction are acquired. The last paragraph adapts the acquired four-point-row proof with its ten actual fixed points stated explicitly. The new step is the exhaustive remaining fixed-point budget. Combined with an independently reviewed lower bound $f \geq 9$, this would exclude the whole single-orbit assignment configuration; that combination is not a premise here. Split assignments and their source routing remain. No new induction, computation, formal, original or question credit is claimed.